

EM 659: Advanced Lean Systems

Course Syllabus

Course Description A class in advanced lean practices as applied in manufacturing and the office. Focus will be on the advanced practical aspects on implementing and maintaining enterprise systems capable of continuous improvement. In-depth treatment will be given value stream analysis as defined by customers, lean logistics, lean cost accounting, zero defect quality practices, and the integration of human resource practices within the enterprise strategy.

Course Objectives EM659 Advanced Lean Systems explores advanced lean practices as applied in manufacturing. This is the third course in Lean Track for all Graduate Engineering Management students at Eastern Michigan University. The course explores lean production systems and focus on the practical aspects on implementing and maintaining production systems capable of continuous improvement. This course will help students to understand the mathematical models utilized in Lean production systems and acts as a first preparatory course in SME Bronze Lean Certification.

At the completion of this course you should be able to:

- A. Trace the evolution of enterprise systems from craft to mass to lean forms.
- B. Articulate the advantages and disadvantages of the lean form of enterprise organization both in a vertically integrated and a de-verticalized global business environment.
- C. Demonstrate understanding of the general elements of lean and their inter-relationship.
- D. Describe the general social and technical issues of implementing lean enterprise in both brownfield and greenfield environments
- F. Provide examples of lean quality techniques and interventions which can lead to zero defects
- G. Describe how the lean business strategy integrates human and technical resources
- H. Develop a rationale for the integration of total productive maintenance (TPM) into lean enterprise.
- I. Be able to describe and conduct a lean transformation analysis in a real-world setting
- E. Demonstrate a basic mastery of conventional mass enterprise cost accounting as well as lean cost accounting and activity-based systems.

Course Policies and Procedures

1. Students are responsible for monitoring messages on opening page several times a week.
2. Students are held responsible for reading and other assignments, tests, information, and materials dispensed during the course.
3. No time extensions will be given for assignments unless agreed upon by the instructor. Late assignments and tests are subject to a reduced grade. Assignments submitted within a week will receive a maximum of one-half the value. Assignments submitted more than a week late will receive a grade of zero.
4. "I"(s) will be given only in accordance with Eastern Michigan University's guidelines.
5. Feedback concerning course requirements, assignments, tests, or content is welcome and encouraged.
6. Students experiencing difficulty with course content should seek a private consultation with

the instructor to discuss remediation of deficiencies.

7. Academic Honesty: Please review the university policy on academic honesty and plagiarism at <http://www.emich.edu/sjs/conductcode.html>

Eastern Michigan University Conduct Code and Judicial Structure for Students and Student Organizations

I.A.11. Academic dishonesty

1.Students are not to engage in any form of academic dishonesty including, but not limited to, plagiarism, alteration of records, substitution of another's work representing it as the student's own, and knowingly assisting another student in engaging in any such activity. For purposes of this section, plagiarism is defined as the knowing use, without appropriate approval, of published materials, expressions, or works of another with intent to represent the material(s) as one's own.

Grading Policy Grades will be determined after a statistical analysis of the class performance on all assignments. Letter grades will be granted based on the criteria listed in the Graduate Catalog.

10	Unit Exams
35	Individual Assignments
25	Team Assignment
15	Threaded Response
15	Final Exam